

EFF-Dock: Separating Candidate Generation, Physical Validity and Selection in Fragment-Based Flow-Matching Docking

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Abstract

Protein–ligand docking must generate, validate and select a near-native pose, yet benchmarks usually report only the selected pose. We present EFF-Dock, a supplied-pocket method that learns an SE(3)-equivariant flow over rigid ligand fragments, with energy refinement and confidence ranking. Across five redocking cohorts, refined 100-pose banks contained an RMSD < 2 Å candidate for 87.9–96.5% of complexes, and selected poses reached 53–80% RMSD–PoseBusters success, not clearly different from the strongest baselines on PoseBusters v2 but below SigmaDock on Astex. PoseBusters checks on all 235,800 Astex and PoseBusters v2 candidates showed that refinement raised candidate validity from 24–27% to 88%, whereas a valid near-native candidate went unselected for 14% of complexes (66–71% of remaining failures). A matched ablation showed no clear advantage of a least-squares Newton–Euler readout over direct prediction. Selection, not generation, was the largest remaining failure source.

A ligand’s three-dimensional pose determines its protein contacts and guides structure-based design. When no experimental complex is available, docking predicts this pose from a receptor structure and a ligand. Here we study the common setting of a fixed receptor with a known binding site.

A docking prediction can fail at three distinct stages. The search may never produce a pose close to the bound one; it may produce such a pose with distorted bonds, inverted stereocentres or steric clashes; or it may produce a valid near-native pose that the scoring function does not rank first. Classical programs combine conformational search with approximate scoring [1, 2], and GNINA adds learned three-dimensional convolutional scoring [3]. Learned generative models now sample poses directly: DiffDock diffuses over translation, rotation and torsions [4], Uni-Mol Docking V2 predicts poses within a specified pocket [5], DiffBindFR adds receptor side-chain flexibility [6], DynamicBind generates ligand-specific receptor conformations [7], FlowDock uses conditional geometric flow matching [8], and AlphaFold 3 predicts complexes by cofolding [9]. PoseBusters showed that many learned methods return poses that are close in RMSD yet physically implausible [10]. A single top-1 success rate cannot show which of the three stages limits a method, and best-of-many metrics mix sampling with selection.

The representation of the ligand shapes all three stages. Rigid-fragment models occupy a middle scale between a rigid ligand and unconstrained atom coordinates: fragments preserve local geometry, whereas torsion-tree parameterizations move a convention-dependent downstream subgraph at each bond rotation. Incremental docking anchors and grows rigid pieces [11], Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) defines retrosynthetically meaningful fragments [12], and SigmaDock applies diffusion to rigid-fragment poses [13]. Flow matching learns a vector field that transports a prior toward the data distribution without integrating the generative ordinary differential equation (ODE) during training [14], and Riemannian formulations accommodate translation–rotation spaces [15, 16]. Equivariant tensor products couple scalar, vector and higher-order features under rigid motion [17], and Newton–Euler mechanics has been used to map interaction energy to flexible-docking motion [18].

Here we present EFF-Dock, a supplied-pocket docking pipeline designed so that each stage can be measured on the same candidates. The ligand is cut at a restricted set of rotatable bonds into rigid

fragments, and an equivariant protein–ligand network learns a flow over fragment translations and rotations. Rather than predicting fragment motion directly, the network predicts one vector per ligand atom, and a least-squares Newton–Euler readout converts these vectors into the rigid motion of each fragment. Generated candidates are refined on a differentiable geometric interaction energy and ranked by a separate confidence network trained for within-bank pose quality [19]. Across five reference-pocket redocking cohorts, we report candidate availability, physical validity and selection on the same banks, apply official PoseBusters checks to every Astex and PoseBusters v2 candidate, compare the Newton–Euler readout with a direct fragment readout under matched training, and compare supplied-pocket docking methods directly while reporting cofolding references with their distinct input conditions.

1 Results

1.1 Fragment-space docking pipeline

EFF-Dock represents a ligand as rigid components obtained by cutting a restricted set of rotatable bonds. A heterogeneous protein–ligand graph conditions a flow over the fragments’ translations and rotations (Fig. 1). An equivariant network predicts one vector per ligand atom; a least-squares rigid-motion projection converts these vectors to fragment velocities (Fig. 2). A separate network scores candidates for ranking within the set generated for each complex, termed a candidate bank. Generated candidates are refined by bounded descent on a differentiable geometric interaction energy and then ranked again. The energy supplies coordinate gradients independently of the learned vector field; it is neither an affinity predictor nor a free-energy model.

Evaluation uses a fixed holo receptor, a reference-derived pocket, 100 candidates and ten integration steps, with three repeats of the same trained weights. On a single RTX PRO 6000 GPU, the complete primary protocol took about 100 s per Astex complex: 8.8 s to prepare the ligand and generate 100 candidates (amortized over one 85-complex process, including model loading), 81.7 s for refinement on the GPU and 9.5 s for confidence scoring of the raw and refined banks, with generation and the per-complex stages timed under the conventions given in Methods. Online Methods and Supplementary Information, Secs. S1.2–S1.3 specify the evaluated coordinate-to-feature conventions and their differences from training.

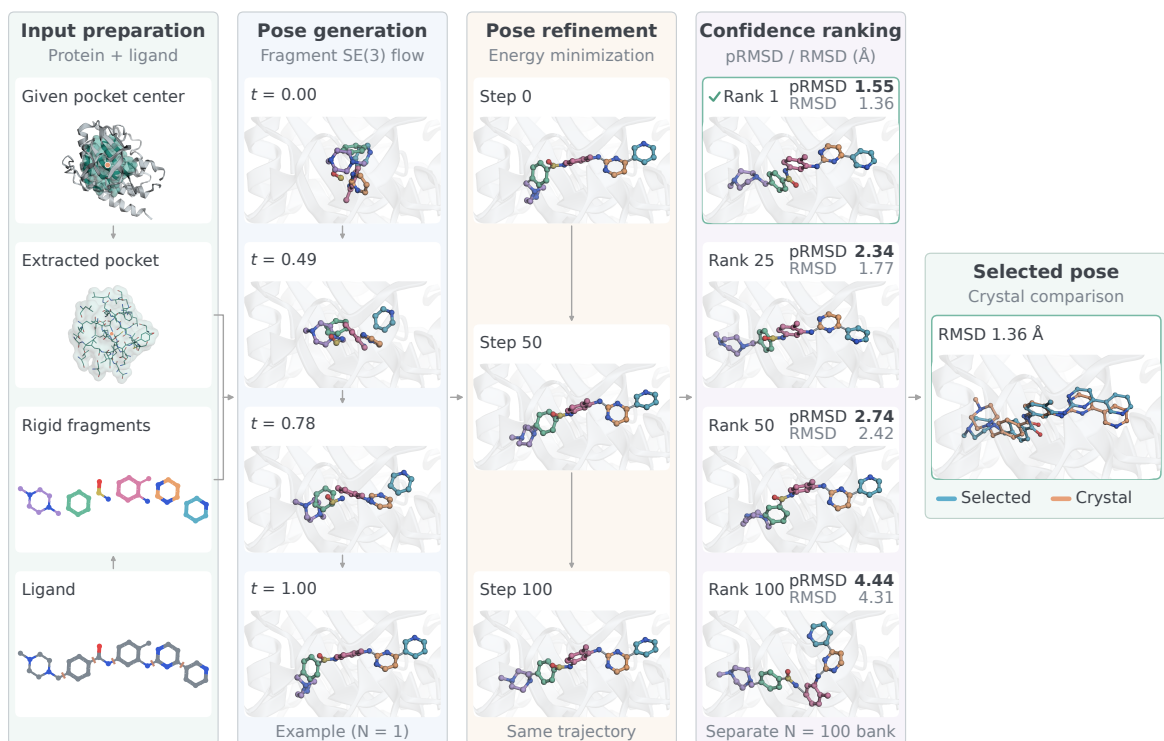
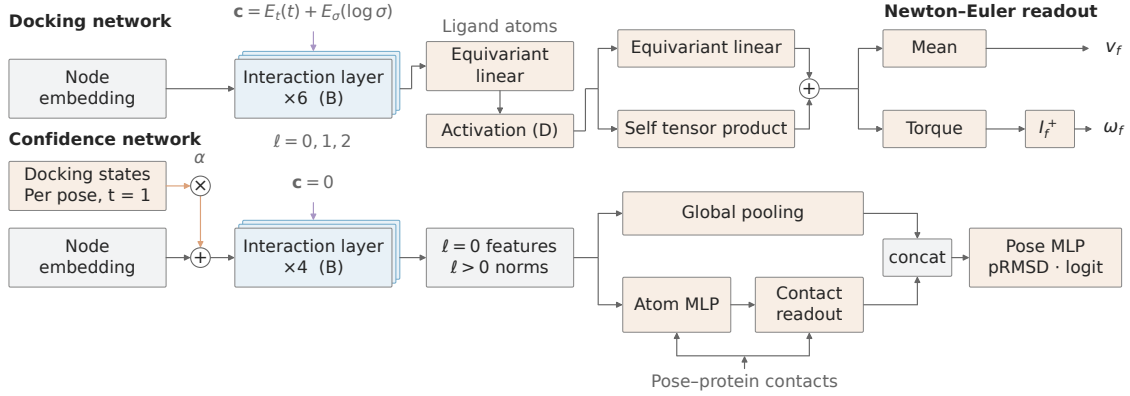
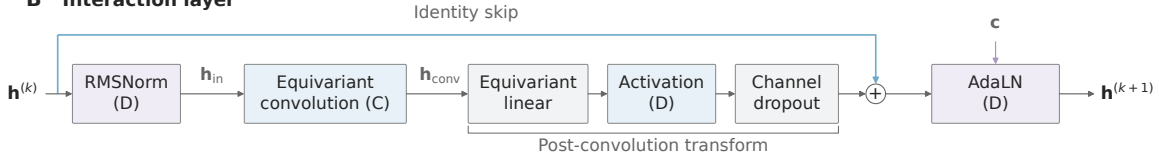


Figure 1: **Overview of the EFF-Dock pipeline, illustrated on Astex complex 1T46.** Input: a prepared receptor, the ligand and a supplied pocket center; the ligand is cut into rigid fragments (colors). Pose generation: fragments move along the learned SE(3) flow from a random prior ($t = 0$) to a predicted pose ($t = 1$); one prespecified trajectory (seed 42) from the supplied conformer is shown, whereas the benchmark bank uses a conformer generated from SMILES. Pose refinement: bounded descent on the geometric interaction energy for the same pose. Confidence ranking: candidates from the separately generated 100-pose benchmark bank are ordered by predicted RMSD (pRMSD), and the top-ranked candidate (check mark) is selected. RMSD values and crystal overlays (orange) are retrospective and are not available to the method; the 1.36 Å selected-pose RMSD belongs to the ranked bank, not to the illustrated trajectory.

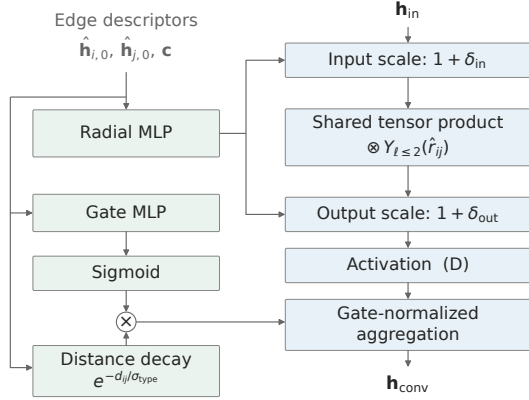
A Model architecture



B Interaction layer



C Equivariant convolution



D Normalization and activation

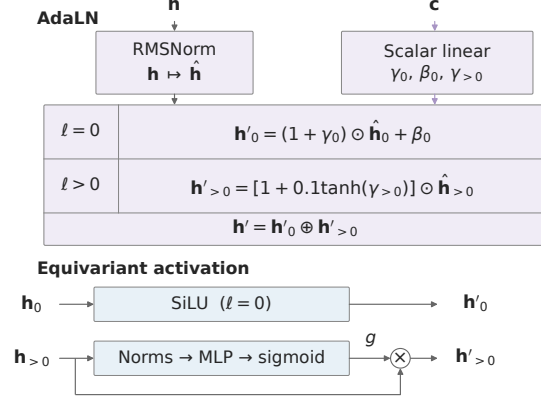


Figure 2: **Docking and confidence architectures.** (A) The docking network applies six conditioned equivariant interaction layers and converts ligand-atom features to fragment translation and angular velocities with the Newton-Euler readout. For each candidate, terminal docking features are recomputed at $t = 1$ using the translation-prior standard deviation used to generate it. The confidence network combines these with a new embedding, applies four interaction layers with zero conditioning vector $c = 0$, converts the resulting features into rotation-invariant quantities, and predicts pose-level RMSD and success scores from global and contact-aware summaries. (B) Each interaction layer contains normalization, equivariant convolution, post-convolution transformation, a residual connection, and adaptive layer normalization. (C) The equivariant convolution uses radial and gating networks, shared tensor products through $\ell \leq 2$, distance decay, and gate-normalized aggregation with invariant nonscalar rescaling. (D) Scalar and nonscalar channels use type-appropriate adaptive normalization and equivariant activation. Detailed tensor definitions are provided in Supplementary Information, Sec. S3.

1.2 Supplied-pocket comparison

Fig. 3 compares selected-pose RMSD success and RMSD–PoseBusters success on Astex Diverse and PoseBusters v2. The local comparison includes SigmaDock [13], DiffDock-Pocket [20], RLDiff RL++ [21], DiffBindFR [6], and force-optimized SurfDock [22]. On Astex, EFF-Dock achieved 80.0% same-pose RMSD–PoseBusters success. On Astex, SigmaDock, SurfDock and RLDiff RL++ had higher point estimates on both endpoints. On PoseBusters v2, EFF-Dock reached 79.1% RMSD–PoseBusters success. Relative to SigmaDock, the paired difference in RMSD–PoseBusters success was +2.7 percentage points (95% bootstrap interval, -2.3 to $+7.5$), compared with -9.0 points (-16.1 to -2.0) on Astex. The PoseBusters interval includes zero, leaving relative performance unresolved; intervals are exploratory and unadjusted for multiple comparisons (Supplementary Fig. S14). Of the selected RMSD-successful EFF-Dock poses, 96.2% on Astex and 97.1% on PoseBusters v2 also passed PoseBusters. On Astex, the paired differences from SurfDock (-5.9 points; -14.9 to $+3.1$) and RLDiff RL++ (-2.4 ; -10.2 to $+5.1$) included zero, as did the PoseBusters v2 difference from SurfDock ($+1.3$; -3.8 to $+6.5$). The local baselines generate 40 poses, whereas the primary EFF-Dock protocol generates 100. A separately generated 40-pose, 25-step EFF-Dock bank, with the same product of candidate count and integration-step count (1,000 pose-steps; the baselines use 800 to 1,200), reached RMSD–PoseBusters success of 78.0% on Astex and 76.7% on PoseBusters v2 (Supplementary Fig. S19): between SigmaDock (76.4%) and SurfDock (77.8%) on PoseBusters v2, and below SigmaDock, SurfDock and RLDiff RL++ on Astex.

Learned methods retain their native preparation, sampling and ranking procedures; GOLD and Vina are literature values [10]. Protocol and candidate-budget differences are specified in Supplementary Information, Sec. S7.4.

Supplementary Table S15 and Fig. S20 extend the comparison to PhiBench-derived, FoldBench and the official 802-case OpenBind subset using benchmark-specific endpoints; on OpenBind, RMSD–PoseBusters success was 53.2% on the 925-case primary cohort and 56.4% on the 802-case subset. They include pocket-conditioned baselines, cofolding references and GNINA holo redocking, with task, cohort, preparation and evaluability differences specified in Sec. S7.3.

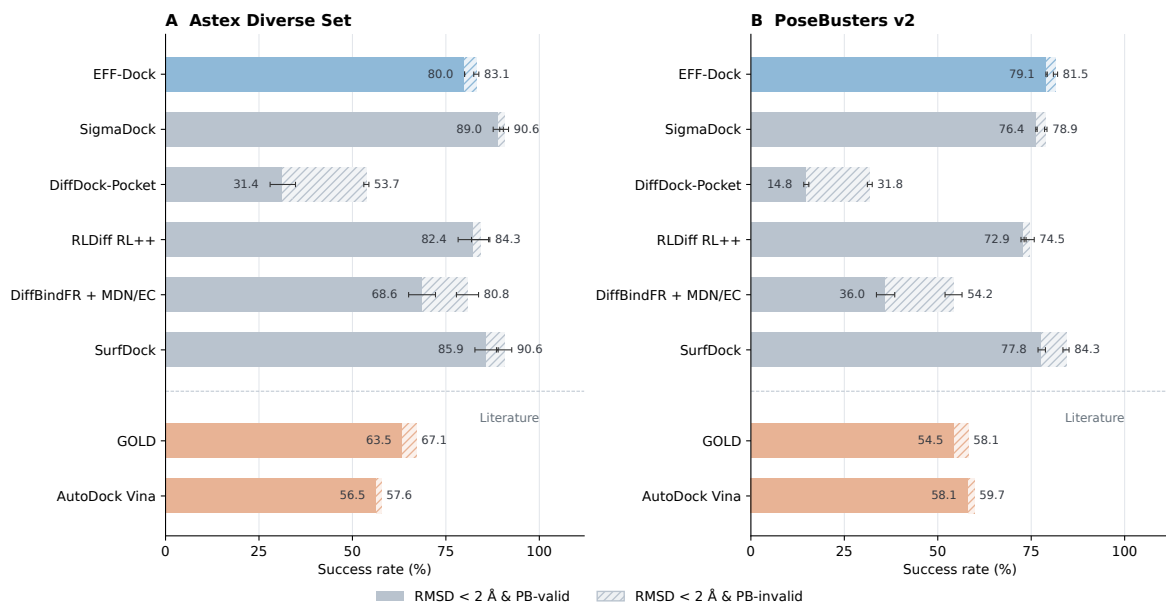


Figure 3: **External docking performance on supplied-pocket benchmarks.** (A) Astex Diverse Set (85 complexes) and (B) PoseBusters v2 (308 complexes). Solid bars show RMSD < 2 Å and PoseBusters success; hatched segments complete RMSD-only success. Local results are means with sample SD across three repeats; EFF-Dock repeats share one starting conformer, so their SD excludes conformer variation. EFF-Dock uses 100 poses, 10 sampling steps, refinement, and chirality-filtered confidence selection. GOLD and AutoDock Vina are literature values; the remaining methods use their native sampling and ranking protocols. DiffDock-Pocket uses a holo-aligned predicted receptor; DiffBindFR samples pocket side-chain torsions on a fixed backbone and uses cognate-ligand coordinates in its upstream scoring graph; its selected ligand poses are evaluated against the common frozen holo receptor. Budgets, receptor preparation, selectors and evaluation versions differ, so these are protocol-conditioned comparisons.

1.3 Newton–Euler readout ablation

To test whether the least-squares Newton–Euler readout contributes to candidate generation, we trained two docking models from scratch that differed only in the output head: the Newton–Euler readout, and a direct readout that predicts each fragment’s translational and angular velocity from its node features (Methods). Under this matched, reduced budget, about 2% of the training samples seen by the released model, the two readouts performed similarly throughout training (Supplementary Fig. S21). At the terminal checkpoint, internal-validation rollout success was 11.1% with the Newton–Euler readout and 10.8% with the direct readout (paired difference +0.3 percentage points; 95% interval -0.8 to $+1.4$). In unrefined 100-pose banks, oracle coverage was 89.4% versus 91.8% on Astex (-2.4 points; -5.9 to 0.0) and 88.3% versus 87.0% on PoseBusters v2 ($+1.3$ points; -1.0 to $+3.9$). The fraction of near-native candidates per bank differed by $+0.5$ percentage points on Astex (-0.3 to $+1.3$) and $+0.2$ on PoseBusters v2 (-0.2 to $+0.6$), and median minimum RMSD was 1.06 versus 0.99 Å on Astex and 1.13 Å for both readouts on PoseBusters v2. Both reduced-budget models remained below the released model, whose unrefined banks reached 96.9% and 94.6% oracle coverage. We found no clear advantage for the least-squares readout under this protocol. The experiment does not establish equivalence and does not assess production-scale training, the ranked pipeline or variability across training seeds.

1.4 Refinement and chirality-aware selection

Fig. 4 compares refinement and chirality-aware selection. Most of the gain from refinement was in physical validity. Under the primary selector, the fraction of selected poses passing all PoseBusters checks rose from 57.1–72.2% for raw banks to 93.7–99.7% for refined banks, whereas RMSD-only success rose by 2.0–8.3 percentage points. RMSD–PoseBusters success therefore increased by 15.7–23.2 points under the primary selector and by 21.7–26.5 points without chirality filtering. These selected-pose gains combine coordinate refinement with confidence reselection.

Chirality filtering added smaller mean improvements of 0.3–2.4 percentage points after refinement (Supplementary Fig. S9). Refinement also increased the fraction of candidates whose tetrahedral centers matched the input stereochemistry from 58.5–74.4% to 88.2–96.8% across cohorts; such centers can be inverted where a cut bond joins two independently moving fragments.

Supplementary Fig. S1 examines how docking outcomes vary with molecular complexity. Selected-pose RMSD success and RMSD–PoseBusters success generally decreased for larger ligands, more rotatable bonds, and more fragments, most clearly in PoseBusters, PhiBench-derived, and FoldBench. Oracle success was substantially less sensitive than selected top-1, so the top-1–oracle gap tended to widen in harder strata. These patterns suggest that molecular complexity is associated with selection loss as well as near-native candidate availability. The smallest Astex and high-complexity bins are too sparse for a clear trend.

Supplementary Fig. S16 resolves the aggregate validity gain into individual PoseBusters checks. Refinement reduced selected-pose failures in bond geometry, internal clashes, receptor clashes and internal energy, whereas stereochemistry and ring-planarity failures changed less consistently. Because refinement can change which candidate is selected, the figure also follows the same candidate before and after refinement, separating coordinate repair from reselection.

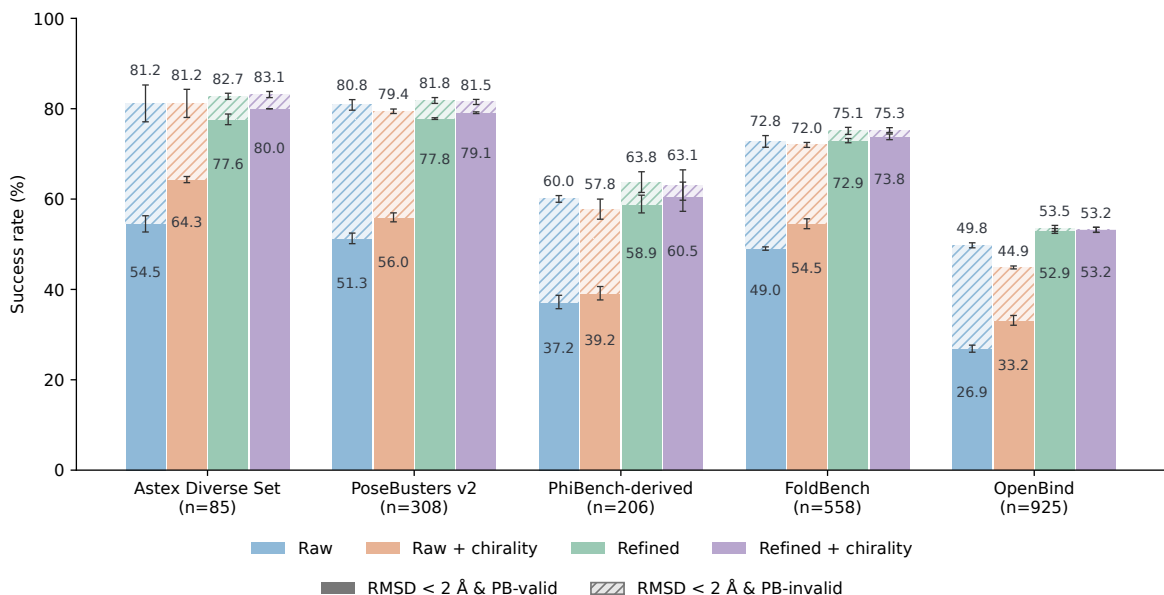


Figure 4: **Effects of refinement and chirality-aware selection.** EFF-Dock results are shown for Astex, PoseBusters v2, PhiBench-derived, FoldBench, and OpenBind using the frozen released weights and unguided 100-pose, 10-step sampling. Raw and refined banks are ranked with or without the chirality filter. Solid bars show RMSD and PoseBusters success; hatched segments complete RMSD-only success. Bars and error bars are the mean and sample SD across three repeats on fixed cohorts; for Astex and PoseBusters v2 the repeats share one starting conformer, whereas for the other cohorts the conformer varies with the repeat.

1.5 Candidate validity and selection headroom

Official PoseBusters checks on every raw and refined candidate in the Astex and PoseBusters v2 banks (235,800 poses) separate coordinate repair from reselection (Supplementary Tables S20 and S21). Only 27.4% and 23.8% of raw candidates passed all checks, including 42.7% and 37.0% of RMSD-successful candidates. After refinement, 88.3% and 88.1% of all candidates and 93.8% and 93.7% of RMSD-successful candidates passed. Following each candidate through refinement, 61.2% and 64.3% changed from failing to passing PoseBusters and 0.4% and 0.1% from passing to failing, whereas 2.6% and 2.8% became RMSD-successful and 1.1% and 1.1% lost RMSD success. Refinement therefore substantially increased PoseBusters validity, whereas relatively few candidates crossed the RMSD-success threshold in either direction. As a result, the joint oracle, the fraction of complexes whose bank contained a candidate that was both RMSD-successful and PoseBusters-valid, rose from 87.5% to 94.1% on Astex and from 78.2% to 93.1% on PoseBusters v2 (Fig. 5D).

Fig. 5 evaluates the ranker within these banks. For refined banks, mean within-complex Spearman correlation between predicted and observed RMSD ranged from 0.51 to 0.70 across cohorts, near-native classification AUROC from 0.88 to 0.94 and average precision from 0.61 to 0.82. Despite this discrimination, selection lost many available successes. Fig. 5C partitions each primary outcome by whether the refined bank lacked an RMSD-successful candidate, all such candidates were excluded by the chirality screen, the selector ranked another candidate first, or the selected RMSD-successful pose failed PoseBusters. Chirality exclusion accounted for at most 0.3% of complexes, whereas ranking failure, defined by RMSD alone, accounted for 12.4% on PoseBusters v2 to 34.4% on OpenBind. With the all-candidate labels, a chirality-eligible candidate that was both RMSD-successful and valid existed but was not selected for 14.1% of Astex and 13.7% of PoseBusters v2 complexes; 5.9% and 6.9% lacked any jointly successful candidate, and chirality screening excluded all such candidates for a further 0.0% and 0.2%. Retrospectively restricting selection to candidates that pass PoseBusters increased selected success only from 80.0% to 80.8% and from 79.1% to 81.0%, leaving most of this headroom unrecovered. Given an eligible jointly successful candidate, the selector chose one for 85.0% and 85.2% of complexes. The joint oracle measures headroom for better selection within these banks; it does not show that a

ranker can recover it.

Ranking loss was largest where near-native candidates were sparse. Only 6.3% of refined OpenBind candidates were RMSD-successful, yet 87.9% of complexes contained at least one. Confidence ranking enriched near-native poses at early ranks relative to generation order (Supplementary Fig. S2), and expanding from top-1 to top-5 increased RMSD coverage by 7.5–20.3 percentage points (Supplementary Fig. S10). In nested prefixes of each bank, oracle coverage never decreased as candidates were added, but selected top-1 success could decline when the ranker promoted an incorrect new pose (Supplementary Fig. S3). Selection regret and reliability are reported in Supplementary Figs. S18 and S12.

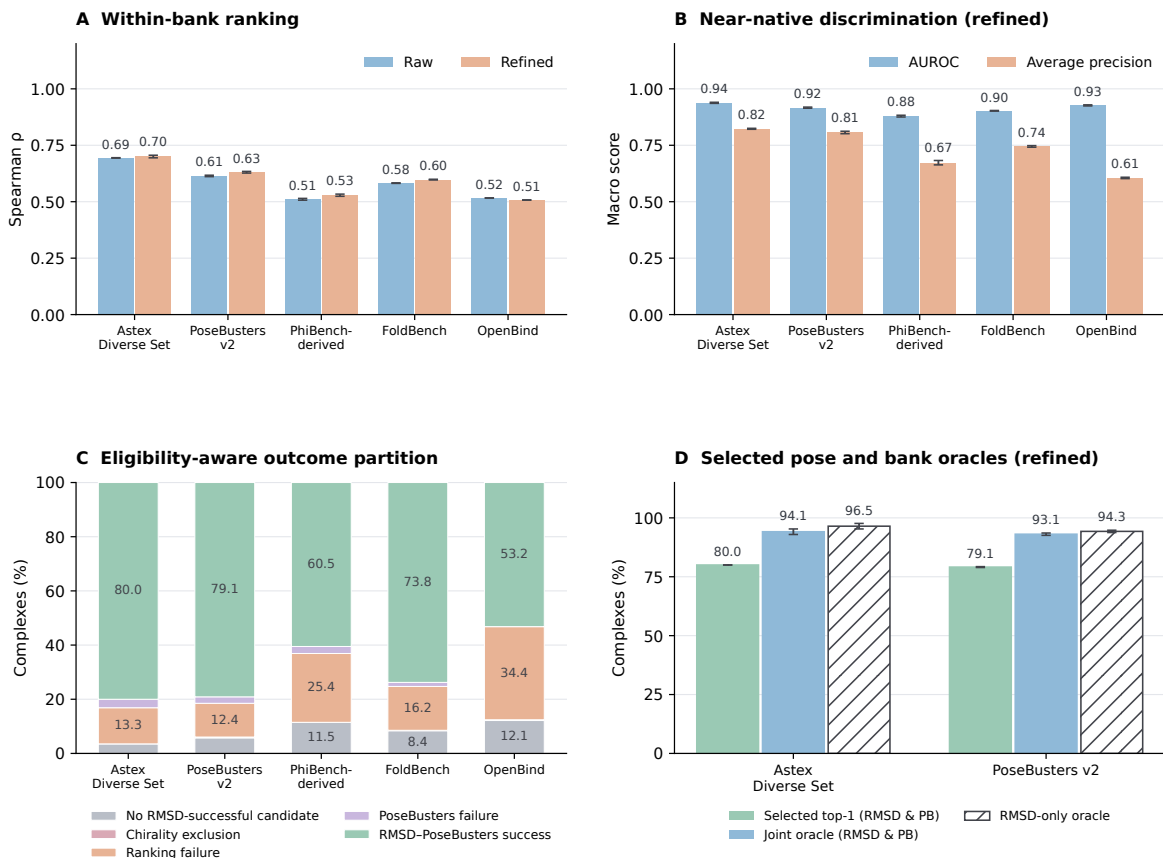


Figure 5: Confidence ranking and selection headroom. (A) Within-complex Spearman correlation between predicted and observed RMSD for raw and refined banks. (B) Refined-bank AUROC and average precision for RMSD < 2 Å labels; banks containing only one class are excluded. (C) Partition of refined primary outcomes: no RMSD-successful candidate in the bank; all such candidates excluded by the chirality screen; an eligible RMSD-successful candidate not selected (ranking failure); the selected RMSD-successful pose fails PoseBusters; or RMSD–PoseBusters success. Candidate success in C is defined by RMSD alone; the corresponding partition with all-candidate PoseBusters labels for Astex and PoseBusters v2 is given in Supplementary Table S21. (D) For Astex and PoseBusters v2, selected-pose RMSD–PoseBusters success compared with the joint oracle (any candidate with RMSD < 2 Å passing all PoseBusters checks) and the RMSD-only oracle, from official checks on every refined candidate. Bars show three-repeat means and sample SD.

1.6 Training-set relatedness

Sequence and ligand overlap with the training set differed substantially among cohorts (Supplementary Figs. S6–S8). Neither exact ligand matches nor sequence-identity strata showed a consistent association with docking success. Performance was nevertheless lower on the most distant complexes. Only 22

external complexes satisfied the simultaneous restrictions of binding-chain identity below 30%, nearest training-ligand Tanimoto similarity below 0.5 and no exact ligand match (Supplementary Fig. S13): 2 from Astex, 4 from PoseBusters v2, 7 PhiBench-derived, 9 FoldBench and none from OpenBind. On the 7 PhiBench-derived and 9 FoldBench complexes in this subset, RMSD-PoseBusters success was 47.6% (10 of 21 complex-repeats) and 55.6% (15 of 27), compared with 60.5% and 73.8% on the full cohorts. On FoldBench, interfaces without a shared training PDB accession reached 69.7%, compared with 76.3% for shared accessions, and the 66 interfaces released after June 2024 reached 68.2% (Supplementary Table S17). These small subsets show lower observed success but do not establish a systematic performance gap attributable to training-set dissimilarity.

1.7 Sensitivity to generation inputs

We varied the generation crop, supplied-center error and translation-prior scale on all Astex and PoseBusters complexes while holding the networks, downstream spatial context and primary selector fixed (Fig. 6). With a 10 Å crop and prior scale 2 Å, RMSD-PoseBusters success was 80.0% on Astex and 79.1% on PoseBusters without center error. Center error with standard deviation 2 Å per axis reduced success to 71.0% and 64.4%, respectively. Refined-bank RMSD-only oracle coverage at 2 Å error was 86.3% and 82.8%.

Larger generation crops partly buffered center error. The prespecified prior scale of 2 Å was not the best setting for an exact center: a 1 Å prior reached 83.9% on Astex and 79.8% on PoseBusters v2. The narrower prior was, however, less tolerant of center error, falling to 65.1% and 58.4% at 2 Å jitter, compared with 71.0% and 64.4% for the 2 Å prior. Fig. 6 reports the complete grids; the production setting was fixed before these grids were measured and was not changed.

Refinement and confidence scoring retained the original unperturbed center in all conditions, with the refinement energy shell and the 10 Å scoring crop unchanged. The crop and center-jitter grids perturb sampling inputs with fixed downstream spatial context. Prior-scale comparisons include the scale’s influence on both sampling and the recomputed docking features used for ranking. Guidance and candidate-allocation controls at equal learned pose-step counts are reported separately (Supplementary Fig. S19).

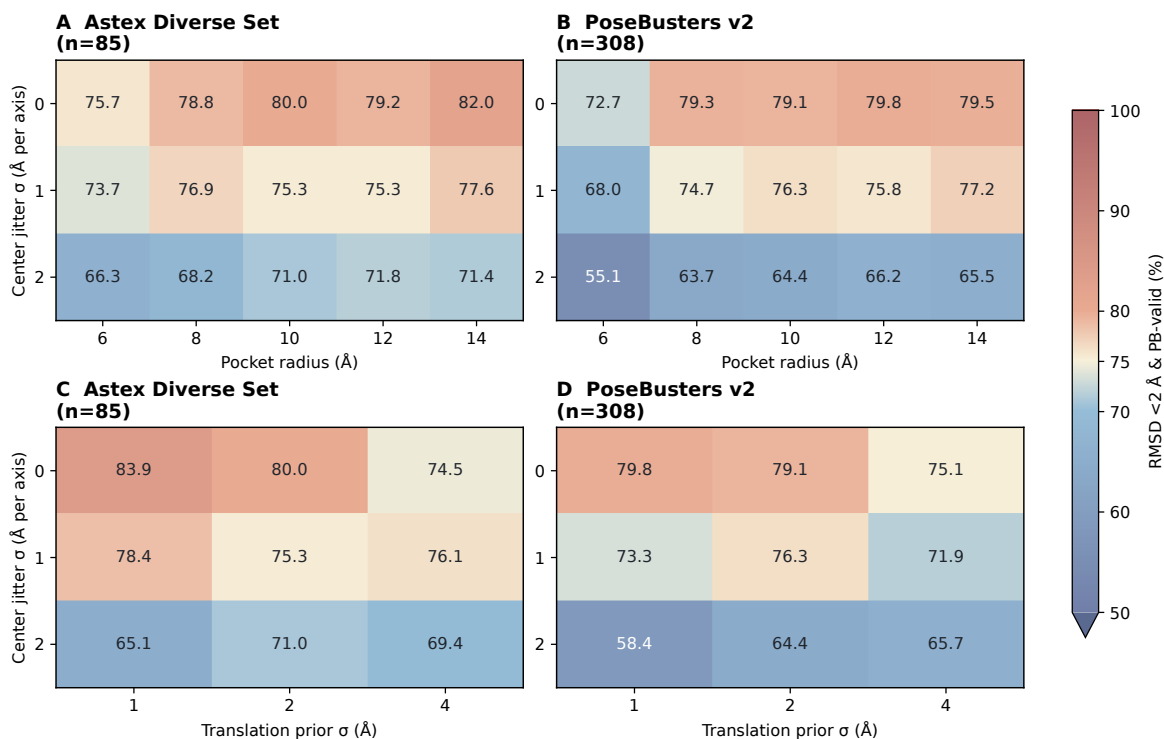


Figure 6: Sensitivity to supplied-pocket radius, center error and translation-prior scale. (A,B) Joint selected-pose RMSD < 2 Å and PoseBusters success across generation crop radii at prior scale 2 Å, for Astex Diverse Set (85 complexes) and PoseBusters v2 (308 complexes). (C,D) Prior-scale dependence at a 10 Å generation crop on the same cohorts. Rows vary Gaussian center error, with standard deviation reported per Cartesian axis. Cells show three-inference-seed means on a linear color scale from 50% to 100%, with the neutral color at 75% and darker blue below 50%. All conditions use unguided 100-pose, ten-step generation, energy refinement and chirality-filtered confidence ranking. Refinement and ranking retain the original center, the refinement energy shell and the 10 Å scoring crop. Terminal docking features used for confidence are recomputed with the translation-prior scale used to generate each candidate. The grid isolates the crop and prior axes rather than crossing every pair. Downstream spatial context is fixed; prior-scale effects include sampling and terminal feature extraction.

2 Discussion

Measuring candidate generation, coordinate validity and selection on the same banks locates where this docking pipeline loses accuracy. Absence of an RMSD-successful candidate was a minority failure mode in the evaluated refined banks: refined 100-pose banks contained an RMSD-successful candidate for 87.9% (OpenBind) to 96.5% (Astex) of complexes, and on Astex and PoseBusters v2 a candidate that was also physically valid for 94.1% and 93.1%. Refinement accounted for most of the observed PoseBusters-validity gain. About one quarter of raw candidates passed PoseBusters, whereas bounded refinement made 88% pass, with relatively few candidates crossing the RMSD-success threshold in either direction. Selection accounted for most of the remaining failures, 71% and 66% of failures on Astex and PoseBusters v2: a valid near-native candidate was available but not selected for about one in seven Astex and PoseBusters v2 complexes, roughly twice the fraction that lacked one.

The readout ablation qualifies the architectural interpretation. A direct fragment readout performed similarly to the Newton–Euler readout under equal reduced-budget training, so the accuracy reported here should be attributed to the fragment flow, refinement and ranking pipeline as a whole rather than to the least-squares readout. The readout remains a convenient construction: it acts on per-atom vectors, assigns no angular velocity to unobservable rotation axes, and gives each fragment velocity an explicit rigid-body interpretation, but these properties did not translate into measurably better candidates in this experiment.

Two features of the ranking loss indicate where improvement is likely. Loss was largest where near-native candidates were sparse, as on OpenBind, and for larger, more flexible ligands, whose banks retained high oracle coverage. In nested generation prefixes, adding candidates never reduced oracle coverage but could reduce top-1 success when the ranker promoted an incorrect new pose. Confidence models for docking should therefore be evaluated not only by discrimination within a fixed bank but also by their stability as the bank grows, and the joint oracle gives the headroom against which such improvements can be measured.

Several limitations qualify these conclusions. Three repeats with the same trained weights measure inference variability, not variability across training runs. The released weights were trained with a non-covariant fragment-orientation feature and are evaluated with the covariant form; a paired control showed small differences, and the readout ablation used the covariant form throughout. Astex and PoseBusters v2 were used repeatedly during development, PhiBench-derived and FoldBench adapt their source tasks to supplied-pocket redocking, and OpenBind concentrates one protease family, including two noncovalent approximations of covalent systems. Training relatedness is substantial, and only 22 external complexes satisfied all three dissimilarity restrictions, too few to establish performance on unrelated targets. Comparisons with other methods retain their native preparation and candidate budgets and are not compute matched.

The evaluated task is holo redocking into a known pocket with one ligand chemical state and starting conformer. Geometry training used crystal-derived fragment templates, rigid-fragment motion cannot correct errors within a fragment, and systematic sensitivity to the starting conformer was not measured. Receptor flexibility, pocket discovery, solvent rearrangement, chemical-state enumeration and covalent docking are outside the evaluated scope, as are unsupported cofactors and metal clusters. The interaction energy encodes geometric preferences and the confidence score ranks pose quality within a bank; neither predicts binding affinity.

Within these limits, EFF-Dock shows that a compact rigid-fragment flow can supply near-native candidates for most supplied-pocket redocking problems, that bounded refinement substantially increases the fraction passing PoseBusters checks, and that selection accounts for most remaining failures on Astex and PoseBusters v2. Measuring all three stages on the same candidates provides a direct way to attribute future gains in docking accuracy.

3 Online Methods

3.1 Coordinate convention and evaluated weights

The coordinate reconstruction in Eq. (2) uses an active local-to-world rotation, $\mathbf{x} = \mathbf{R}\mathbf{y} + \mathbf{T}$. Equivariance is defined with the prepared fragment-local coordinates and the orientation of their local axes held fixed. Under a joint rigid transformation of the receptor and current ligand pose, the prepared local

coordinates remain fixed while world coordinates transform as $\mathbf{x}' = \mathbf{Q}\mathbf{x} + \mathbf{a}$ and fragment rotations as $\mathbf{R}' = \mathbf{Q}\mathbf{R}$. Supplied conformers are centered fragment-wise without canonicalizing their local axes. Accordingly, a learned fragment-local vector $\mathbf{w}$ is injected into world-frame features as

$$\mathbf{h} = \mathbf{R}\mathbf{w}, \quad \mathbf{h}' = (\mathbf{Q}\mathbf{R})\mathbf{w} = \mathbf{Q}\mathbf{h} \quad (1)$$

under a joint world rotation $\mathbf{Q} \in \text{SO}(3)$. Source-local edge coordinates of the form $\mathbf{R}^\top(\mathbf{x}_j - \mathbf{x}_i)$ perform the inverse world-to-local map.

All evaluations of the released pipeline use fixed docking and confidence weights, without retraining, probability recalibration or result-dependent thresholding. The released weights were trained with a historical injection $\mathbf{R}^\top\mathbf{w}$, which does not transform covariantly under joint rotations. All reported results execute the covariant map of Eq. (1) with the same weights. In a paired control on Astex and PoseBusters v2 with identical seeds and an earlier version of the downstream scoring pipeline, executing the historical map instead changed refined RMSD–PoseBusters success from 80.0% to 78.4% and from 79.0% to 79.4%, respectively; both arms of this control share the earlier conformer and frame handling, so the comparison isolates the injection map. Because the weights were trained with the historical map, the covariant map is an input transformation outside training; the paired control shows a small change in aggregate success under the tested conditions but does not establish weak reliance on this feature or interchangeability of the two maps. Under the covariant map, joint rotations of the receptor and candidate changed network outputs by at most 1.8×10^{-5} in relative norm, and the frame convention for fragments with an unobservable twist changed predicted RMSD by at most 1.5×10^{-3} Å. Coordinate-to-feature conventions and the numerical checks are documented in the [Supplementary Information, Sec. S1.2](#); the readout ablation was trained with the covariant map throughout.

3.2 Task definition and fragment pose representation

EFF-Dock takes a prepared fixed receptor $\mathcal{P}$, a heavy-atom ligand graph $\mathcal{G}_L$ in one specified chemical state, one supplied or generated ligand conformer $\mathcal{C}_L$, and an explicit pocket center $\mathbf{b} \in \mathbb{R}^3$. We denote their complete conditioning context by $\mathcal{C}_{\text{cond}} = (\mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \mathbf{b})$. It returns a finite set of ligand poses and a separate score used only to rank poses within that set. Deployment inputs exclude target ligand coordinates, target-derived frames, RMSD, validity outcomes and benchmark identity. The reference-pocket experiments derive the supplied center from the reference structure before removing the ligand.

The ligand contains N heavy atoms partitioned into K rigid fragments, with $i \in \{1, \dots, N\}$ and $k \in \{1, \dots, K\}$ indexing atoms and fragments, respectively. The map $g(i)$ assigns atom i to a fragment, and $\mathbf{y}_i \in \mathbb{R}^3$ is its local coordinate relative to that fragment’s geometric centroid. The generative state and atom reconstruction are

$$\mathcal{M}_L = \prod_{k=1}^K (\mathbb{R}^3 \times \text{SO}(3)), \quad \mathbf{z} = \{(\mathbf{T}_k, \mathbf{R}_k)\}_{k=1}^K, \quad \mathbf{x}_i(\mathbf{z}) = \mathbf{R}_{g(i)}\mathbf{y}_i + \mathbf{T}_{g(i)}. \quad (2)$$

Here $\mathcal{M}_L$ is the ligand pose space, $\mathbb{R}^3$ is the translation space, and $\text{SO}(3)$ is the group of proper three-dimensional rotations. The group $\text{SE}(3)$ combines proper rotations and translations. Coordinates and distances use Angstrom, angles use radians unless degrees are shown, and flow time is dimensionless. Superscript $\top$ denotes transpose and $\mathbf{I}_3$ is the three-dimensional identity matrix; further notation and parameter conventions are collected in the [Supplementary Information, Sec. S1.1](#). For fragment k , $\mathbf{T}_k \in \mathbb{R}^3$ is the position of its geometric centroid and $\mathbf{R}_k \in \text{SO}(3)$ maps its local coordinates into the pocket-centered frame; $\mathbf{z}$ collects all fragment states, and $\mathbf{x}_i(\mathbf{z}) \in \mathbb{R}^3$ is the reconstructed coordinate of ligand atom i in the pocket-centered frame. This representation preserves intrafragment geometry while allowing fragments to translate and rotate relative to one another.

Ligands are standardized with RDKit [23]: stereochemistry is assigned before hydrogen removal, the largest connected component is retained, and formal charge and the supplied protonation state are preserved. For simplified molecular-input line-entry system (SMILES) input, the inference path constructs one seeded Experimental-Torsion Knowledge Distance Geometry version 3 (ETKDGv3) conformer [24], followed by at most 200 Merck Molecular Force Field (MMFF) iterations [25]; supplied structure-data file (SDF) or Tripos MOL2 coordinates are retained. The receptor is hydrogen-stripped and fixed, and all receptor and ligand coordinates are translated by $-\mathbf{b}$ so that the supplied pocket

center is the origin. For geometry training, each deposited crystal ligand pose is fragmented in its bound conformation, and the resulting crystal coordinates define the fixed fragment-local coordinates used as model input and reconstruction templates. Prospective inputs instead use supplied or independently generated conformers; rigid-fragment motion cannot correct internal differences within a fragment. Exact input normalization and pocket provenance are given in the [Supplementary Information, Sec. S1](#).

3.3 Fragmentation and pocket-conditioned graph

Fragmentation cuts eligible heavy-atom single bonds outside rings while excluding planar carbonyl C–N/O and sulfonyl S–N linkages. Connected components define provisional rigid fragments, singleton components are merged into adjacent fragments, and the retained cut topology supplies fragment adjacency and local cross-fragment distance constraints. This indexed greedy rule is deterministic for a fixed atom ordering; exact eligibility, tie-breaking, and failure handling are specified in the [Supplementary Information, Sec. S2.1](#).

The receptor is cropped residue-wise around the supplied center. Geometry training samples a radius uniformly from 6 to 12 Å and applies isotropic center jitter with standard deviation 2 Å; held-out validation uses an 8 Å crop, and the external reference-pocket protocol uses a fixed 10 Å crop. The graph contains nodes representing ligand atoms, ligand fragments, protein atoms and protein residues; residue nodes are virtual rather than individual atoms. Typed static relations encode covalent structure, fragment membership and topology, protein hierarchy, and residue–fragment context, while protein-atom–ligand-atom contacts within 5 Å are rebuilt at every forward pass. Node chemistry, all ten directed relation types, crop fallbacks, and edge descriptors are listed in the [Supplementary Information, Sec. S3.1](#).

3.4 Equivariant fragment flow and Newton–Euler readout

After pocket centering, fragment translations are drawn from an isotropic Gaussian and multi-atom fragment rotations from the uniform distribution on $\text{SO}(3)$ [26]. Let $t \in [0, 1]$ denote dimensionless flow time, with subscripts 0 and 1 denoting the sampled prior and bound target states, respectively, and let $\sigma > 0$ be the translation-prior standard deviation in Angstrom. For a paired prior and target state, translations follow a linear path and rotations follow the shortest geodesic, with rotation increments applied by left multiplication:

$$\begin{aligned} \mathbf{T}_{k,0} &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3), & \mathbf{R}_{k,0} &\sim \text{Uniform}(\text{SO}(3)), \\ \mathbf{T}_{k,t} &= (1-t)\mathbf{T}_{k,0} + t\mathbf{T}_{k,1}, & \mathbf{R}_{k,t} &= \text{Exp}(t\boldsymbol{\omega}_k^*)\mathbf{R}_{k,0}, \\ \mathbf{v}_k^* &= \mathbf{T}_{k,1} - \mathbf{T}_{k,0}, & \boldsymbol{\omega}_k^* &= \text{Log}(\mathbf{R}_{k,1}\mathbf{R}_{k,0}^{-1}). \end{aligned} \quad (3)$$

Here $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3)$ is the three-dimensional Gaussian prior and $\text{Uniform}(\text{SO}(3))$ is the normalized Haar distribution over rotations. The maps $\text{Exp} : \mathbb{R}^3 \rightarrow \text{SO}(3)$ and $\text{Log} : \text{SO}(3) \rightarrow \mathbb{R}^3$ are the rotation exponential and principal logarithm under the axis-angle identification. The starred vectors $\mathbf{v}_k^*$ and $\boldsymbol{\omega}_k^*$ are the target translational and angular tangents, measured in Angstrom and radians per unit flow time, respectively. Singleton rotations are fixed to identity and contribute no supervised angular mode. At time t , $\mathbf{z}_t = \{(\mathbf{T}_{k,t}, \mathbf{R}_{k,t})\}_{k=1}^K$ denotes the interpolated ligand state. Conditional flow matching trains the vector field $u_\theta(\mathbf{z}_t, t, \mathcal{C}_{\text{cond}}, \sigma)$, with learned parameters θ , to return the fragment-wise predictions $\{(\hat{\mathbf{v}}_k, \hat{\boldsymbol{\omega}}_k)\}_{k=1}^K$ and match the starred targets without integrating the generative ODE during training [14]. The prior-scale mixture, time distribution, rotation augmentation, and numerical path conventions are recorded in the [Supplementary Information, Sec. S2](#).

Equivariant trunk and Newton–Euler readout. All node types share six equivariant interaction layers with 384 scalar channels, 32 channels of each vector parity and 16 channels of each rank-two parity. Time and prior scale enter through a 128-dimensional conditioning embedding. Each layer combines normalized sender features, radial edge features and real spherical harmonics through order two using tensor products, gated residual aggregation and adaptive normalization. Static relations and rebuilt protein–ligand contacts use relation-specific edge features. Complete equations and dimensions are given in the [Supplementary Information, Sec. S3](#).

The output head predicts a velocity proposal $\mathbf{f}_i$ for each ligand atom; this proposal transforms as a polar vector. Its unit-weight Newton–Euler aggregation is a least-squares rigid-body velocity fit

restricted to the retained rotational modes; these vectors are not physical forces or energy gradients. For fragment k , with atom set $\mathcal{A}_k$, center $\mathbf{T}_k$, and lever arm $\mathbf{r}_i = \mathbf{x}_i - \mathbf{T}_k$, the fragment field is obtained by

$$\begin{aligned}\hat{\mathbf{v}}_k &= \frac{1}{|\mathcal{A}_k|} \sum_{i \in \mathcal{A}_k} \mathbf{f}_i, & \boldsymbol{\mu}_k &= \sum_{i \in \mathcal{A}_k} \mathbf{r}_i \times \mathbf{f}_i, \\ \mathbf{J}_k &= \sum_{i \in \mathcal{A}_k} (\|\mathbf{r}_i\|_2^2 \mathbf{I}_3 - \mathbf{r}_i \mathbf{r}_i^\top), & \hat{\boldsymbol{\omega}}_k &= \mathbf{J}_k^+ \boldsymbol{\mu}_k.\end{aligned}\quad (4)$$

Here $\hat{\mathbf{v}}_k$ is the predicted translational tangent, $\boldsymbol{\mu}_k$ is the moment of the atom vectors, $\mathbf{J}_k$ is the unit-mass geometric inertia about the fragment centroid, and $\hat{\boldsymbol{\omega}}_k$ is the predicted angular tangent. The truncated pseudoinverse $\mathbf{J}_k^+$ removes geometric-inertia eigenmodes at or below 1% of the largest eigenvalue, and the same observable subspace is applied to the angular training target. This removes angular-velocity components that are unobservable or poorly determined by the fragment geometry; the exact eigenspace construction is given in the [Supplementary Information, Sec. S3.2](#).

3.5 Objective and sampling

The primary losses are mean-squared errors (MSEs) on fragment translation and observable angular velocity. The geometry objective combines fragment translation and observable angular-velocity errors with two auxiliary terms:

$$\mathcal{L}_{\text{geom}} = \mathcal{L}_v + \lambda_\omega \mathcal{L}_\omega + \lambda_{\text{atom}} \mathcal{L}_{\text{atom}} + \lambda_{\text{DG}} \mathcal{L}_{\text{DG}}. \quad (5)$$

Here $\mathcal{L}_v$ is the MSE in fragment translational velocity, $\mathcal{L}_\omega$ is the MSE in angular velocity after projecting the target onto retained rotational modes, and $\mathcal{L}_{\text{atom}}$ compares reconstructed atom velocities. The distance-geometry term $\mathcal{L}_{\text{DG}}$ compares interfragment atom-pair distances after a one-step endpoint extrapolation, with t^2 weighting and aggregation over valid complexes. Unobservable terms contribute zero. The final continuation uses weights $(\lambda_\omega, \lambda_{\text{atom}}, \lambda_{\text{DG}}) = (8, 0.3, 3)$; definitions, normalizations and distributed reductions are given in the [Supplementary Information, Sec. S3.3](#).

The final continuation samples flow time from an 80% base distribution, a 10% uniform component on $[0, 0.3]$, and a 10% point mass at $t = 0$. The base distribution mixes uniform sampling with a late-biased logit-normal distribution. Its parameters and the prior-scale mixture are specified in the [Supplementary Information, Sec. S2](#).

At inference, sampled fragment priors are integrated from $t = 0$ to 1 using 10 late-biased Euler intervals in the primary protocol. For interval width Δt , translations receive additive increments and rotations are advanced by left multiplication with $\text{Exp}(\Delta t \tilde{\boldsymbol{\omega}}_k)$, where the tilde denotes the guided field when guidance is enabled. Atom coordinates and dynamic protein–ligand contacts are rebuilt before every field evaluation. The sampling schedule and candidate selection are specified in the [Supplementary Information, Sec. S2.2](#).

3.6 Confidence-based pose selection

Pose generation and top-1 selection are separated. For each candidate, fragment translations and rotations are recovered from its atom coordinates. The fixed docking network is then evaluated at $t = 1$, using these fragment states and the translation-prior scale used to generate the candidate, to recompute terminal ligand representations. A second equivariant network combines these with a fresh heterogeneous-graph embedding and applies four additional interaction layers. Invariantized node features and explicit protein-contact summaries are pooled by ligand-contact attention and type-wise mean–maximum aggregation. Atom and pose heads are trained with displacement and RMSD regression, 2 Å success classification, pairwise ranking, and a success-focused listwise objective. For a candidate set $\{\mathbf{z}_S^{(m)}\}_{m=1}^M$, the deployed selector is

$$\mathcal{E} = \begin{cases} \{m : q_m = 1\}, & \text{if any } q_m = 1, \\ \{1, \dots, M\}, & \text{otherwise,} \end{cases} \quad m^* = \arg \min_{m \in \mathcal{E}} c_\phi(\mathbf{z}_S^{(m)}, \mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \sigma). \quad (6)$$

Here $q_m \in \{0, 1\}$ indicates whether candidate m matches all assigned input tetrahedral stereocenters, and $\mathcal{E}$ contains the chirality-matching candidates, or all candidates if none matches. Ties follow saved

candidate order. The unfiltered argmin is reported separately as a diagnostic condition. The confidence network’s nonnegative predicted-RMSD score is c_ϕ , with learned parameters ϕ ; $m^\star$ is the selected candidate index. The prior scale enters this score through the recomputed docking representations. The score ranks candidates within a bank; empirical reliability is evaluated separately from the deployed selector. Energy restraints and confidence features use conformers prepared by the same procedure and seed as pose generation. Refinement preserves each saved candidate’s fragment-internal coordinates. For rescoring, fragment translations and rotations are recovered from candidate coordinates by Kabsch fitting restricted to proper rotations for templates of rank at least two; lower-rank templates use a convention based on input atom ordering and the surrounding geometry. This convention specifies the recovered frames, but network predictions can still depend on rotations of a fragment frame that leave its atom coordinates unchanged; numerical checks and their limits are reported in the [Supplementary Information, Sec. S1.3](#). The complete readout and confidence objective are given in the [Supplementary Information, Sec. S4](#).

3.7 Differentiable guidance and post-sampling refinement

For ligand coordinates $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$, we define a differentiable guidance energy

$$\begin{aligned} E_G &= E_{\text{phys}} + E_{\text{int}}, \\ E_{\text{phys}} &= E_{\text{bond}} + E_{\text{angle}} + E_{\text{proper}} + E_{\text{chiral}} + E_{\text{planar}} \\ &\quad + E_{\text{LJ}} + E_{\text{steric}} + E_{\text{obs}}, \\ E_{\text{int}} &= E_{\text{hyd}} + E_{\text{HB}} + E_{\text{chg}} + E_\pi + E_{\text{cat-}\pi} + E_{\text{XB}} + E_{\text{metal}}. \end{aligned} \quad (7)$$

The cut-interface terms restrain only bond, angle, proper-torsion, chiral, and planar geometry that spans independently moving fragments. The nonbonded terms include a soft-core Universal Force Field (UFF)-style Lennard–Jones profile on admitted interfragment ligand–ligand and protein–ligand pairs, a compact receptor-overlap barrier and repulsion-only obstacles for unresolved receptor chemistry [27]. The seven interaction terms describe hydrophobic contacts, directional hydrogen bonds, screened formal-charge groups, pi stacking, cation–pi contacts, halogen bonds, and typed metal coordination [28, 29, 30, 31, 32]. These terms define a geometric refinement objective in nominal energy units.

Compact radial and angular gates use a clipped quintic smoothstep with zero first and second derivatives at its endpoints. The envelope joins constant regions with continuous energy and first and second derivatives. Energy forms, occupancy aggregation, typing conventions and numerical constants are specified in the [Supplementary Information, Sec. S5](#).

The coordinate gradient $\nabla_{\mathbf{x}_i}$ gives the guidance force $\mathbf{F}_i = -\nabla_{\mathbf{x}_i} E_G$, which is distinct from the learned atom-vector proposal $\mathbf{f}_i$. For atom mass m_i , the fragment mass is $m_k^{\text{frag}} = \sum_{i \in \mathcal{A}_k} m_i$, the center of mass is $\mathbf{c}_k = (m_k^{\text{frag}})^{-1} \sum_{i \in \mathcal{A}_k} m_i \mathbf{x}_i$, and the resultant is $\mathbf{F}_k = \sum_{i \in \mathcal{A}_k} \mathbf{F}_i$. The vector $\boldsymbol{\tau}_k = \sum_{i \in \mathcal{A}_k} (\mathbf{x}_i - \mathbf{c}_k) \times \mathbf{F}_i$ is the center-of-mass torque, and $\mathbf{I}_k$ is the corresponding mass-weighted inertia tensor. The translational and angular guidance directions are

$$\mathbf{d}_k^\omega = \mathbf{I}_k^+ \boldsymbol{\tau}_k, \quad \mathbf{d}_k^T = \frac{\mathbf{F}_k}{m_k^{\text{frag}}} + \mathbf{d}_k^\omega \times (\mathbf{T}_k - \mathbf{c}_k). \quad (8)$$

Here $\mathbf{d}_k^T$ and $\mathbf{d}_k^\omega$ are guidance directions, $\mathbf{I}_k^+$ is the mass-inertia truncated pseudoinverse, and $\mathbf{T}_k - \mathbf{c}_k$ corrects for the offset between the fragment’s geometric-centroid state origin and its center of mass. For guidance experiments, let $(\Delta \mathbf{v}_k, \Delta \boldsymbol{\omega}_k)$ denote the mass–inertia directions after matching the root-mean-square magnitude of the learned atom velocities, increasing guidance strength over late-time integration intervals, and applying caps on translation, rotation and atom displacement. The guided field is

$$\tilde{\mathbf{v}}_k = \hat{\mathbf{v}}_k + \Delta \mathbf{v}_k, \quad \tilde{\boldsymbol{\omega}}_k = \hat{\boldsymbol{\omega}}_k + \Delta \boldsymbol{\omega}_k, \quad (9)$$

with pre-cap guidance strength η scaling $\Delta \mathbf{v}_k$ and $\Delta \boldsymbol{\omega}_k$; the exact interval average, normalization, zero-direction rule, and caps are given in the [Supplementary Information, Sec. S5.9](#). This inference-time modification is conceptually related to guidance of flow-matching vector fields, although the normalization and fragment projection used here are method specific [33]. Post-sampling refinement updates only the fragment state of each candidate. At refinement iteration r ,

$$\mathbf{T}_k^{(r+1)} = \mathbf{T}_k^{(r)} + \alpha_r \bar{\mathbf{d}}_k^T, \quad \mathbf{R}_k^{(r+1)} = \text{Exp}(\alpha_r \bar{\mathbf{d}}_k^\omega) \mathbf{R}_k^{(r)}. \quad (10)$$

Here $(\bar{\mathbf{d}}^T, \bar{\mathbf{d}}^\omega) = \text{Cap}(\mathbf{d}^T, \mathbf{d}^\omega)$ are capped increments and α_r is the accepted line-search step size. The cap first bounds each fragment translation and rotation separately, then jointly rescales both until the reconstructed atom displacement meets its bound. Writing $E_r = E_G(\mathbf{x}^{(r)})$, backtracking reduces α_r until the proposed update yields a finite energy no greater than E_r . The refinement is bounded and stops on convergence, line-search failure, or a sustained energy plateau; the numerical safeguards and termination criteria are given in the [Supplementary Information, Sec. S5.10](#). Raw and refined candidates are compared with unchanged network parameters.

The guidance energy uses a typed 18 Å receptor shell, separate from the 10 Å neural crop. Full typing rules, energy equations, mass-inertia projection, refinement safeguards, and permitted information are provided in the [Supplementary Information, Sec. S5](#).

3.8 Data curation and split

Training data were derived from PLINDER 2024-06/v2 [34]. Curation applies ligand validity, covalency, interaction, experimental-quality, resolution, contact and size criteria, followed by deterministic representative selection within ligand and pocket-community groups. The filters and cohort counts are detailed in the [Supplementary Information, Sec. S6](#).

After model-specific input filtering, docking uses 47,277 training and 1,076 validation samples, while confidence uses 43,092 training and 1,035 validation samples. Their differing input requirements and overlap are detailed in the [Supplementary Information, Sec. S6](#).

External exclusion required both an exact ligand match and membership in a benchmark-associated pocket community, removing 904 samples. Recurring ligands and homologous proteins are assessed separately through training-relatedness analysis.

3.9 Evaluation scope

Row-level splits can share repeated ligands or homologous proteins, so training-set relatedness is measured explicitly rather than inferred from split labels [34, 35, 36, 37]. For each external cohort, we report the receptor state, pocket source, ligand preparation, candidate budget, and selector. Astex and PoseBusters were used during development and are treated as descriptive benchmarks.

The geometry model uses the receptor crop, ligand graph, starting conformer, pocket center, and current sampled coordinates. The confidence model ranks the generated candidates from the same task inputs. The interaction energy may additionally read a larger receptor shell, but it has no access to the crystal pose, RMSD, or validity labels. Raw coordinates are saved before refinement.

3.10 Model training and selection

The docking model was trained for 200,000 base updates, 100,000 geometry-fine-tuning updates and 50,000 early-time-replay updates, with global batch sizes of 256, 256 and 64. The final stage used AdamW with peak learning rate 2×10^{-5} , weight decay 0.01, gradient clipping at 1.0 and exponential moving average (EMA) decay 0.999 [38]; its terminal EMA was evaluated. The confidence model was trained on raw and refined poses and selected at update 70,000 by predicted-RMSD-ranked top-1 success on 1,035 internal-validation complexes. Training stages, initialization, optimization schedules and selection criteria are specified in the [Supplementary Information, Sec. S6.4](#). External benchmark results did not select the evaluated checkpoints.

3.11 Benchmarks and comparison protocol

The evaluation uses five complementary cohorts. The Astex Diverse Set contains 85 curated complexes [39], and PoseBusters v2 contains 308 complexes with detailed coordinate-validity checks [10]. The PhiBench-derived cohort comprises 206 locally selected complexes from the public PhysDock archive; identity with the published 206-case cohort has not been established [40]. FoldBench-Pocket contains 558 released protein-ligand interfaces, including a 66-interface post-June 2024 slice [41]. OpenBind contains 925 EV-A71/CVA16 2A-protease complexes and is an auxiliary single-family cohort, including two flagged noncovalent approximations of covalent systems [42].

All five cohorts are evaluated by holo-receptor, reference-pocket redocking. The pocket center is computed from receptor residue nodes within 8 Å of the reference ligand, and the reference ligand is then removed from the model input.

The primary setting uses 100 candidates, 10 late-biased Euler steps, $\sigma = 2$ Å, a 10 Å receptor crop, and no inference-time guidance. Each cohort uses the same trained weights with inference seeds 42, 100042 and 200042. Astex and PoseBusters fix the ETKDG seed at zero for generation; PhiBench-derived, FoldBench and OpenBind use the per-complex sampling seed for conformer generation, so their repeat variability also includes starting-conformer variation. Candidates are scored before and after bounded refinement. The primary selector removes chirality-mismatched candidates and then chooses the lowest predicted symmetry-aware RMSD; if none passes the chirality screen, it uses the unfiltered ranking. Failed complexes remain in the cohort denominator. Stage costs of the primary protocol were measured on one RTX PRO 6000 Blackwell Max-Q GPU for Astex repeat 0: generation was timed as total wall time of a single 85-complex process divided by 85, including model loading, ligand preparation and output, and refinement and confidence scoring were timed per complex in separate processes, each with one or two CPU threads. Supplementary Figs. S4 and S5 additionally report runtime and memory of the original multi-GPU candidate-acquisition runs, which exclude the later refinement and rescoring passes used for the reported evaluation. Sensitivity to generation inputs is analyzed in Fig. 6; supplementary guidance controls are described below.

3.12 Generation-input sensitivity and guidance controls

Sensitivity experiments use all 85 Astex and 308 PoseBusters complexes with the same three inference seeds and chirality-filtered selector. At translation-prior scale 2 Å, generation crop radii 6, 8, 10, 12, 14 Å are crossed with Gaussian center jitter of 0, 1, 2 Å per Cartesian axis. Separately, prior scales 1, 2, 4 Å are crossed with the same jitter levels at a 10 Å crop. Shared cells are evaluated once, and the primary baseline is reused from the primary evaluation. The jitter random stream is independent of the pose prior, and matching seeds couple prior draws within a candidate budget. Refinement and confidence scoring always use the original center, the refinement energy shell and the 10 Å scoring crop. Docking features used for confidence scoring are recomputed with the translation-prior scale used to generate each candidate.

Four budget/guidance conditions, including the shared primary baseline, cross 100 poses/ten steps or 40 poses/25 steps with unguided sampling or normalized energy guidance of strength $\eta = 2$. Both budgets use 1,000 learned pose-steps, but guidance and candidate processing add work, so equal pose-step counts do not imply equal runtime. Forty- and 100-pose banks are generated separately under prespecified control settings.

3.13 Readout ablation

To test the Newton-Euler readout, two docking models were trained from scratch under the same data, split, configuration, seed, optimizer schedule and hardware type, differing only in the output head. The Newton-Euler arm used the readout of Eq. (4). The direct arm applied the same head structure to fragment-node features and emitted one translational (polar) and one angular (axial) vector per fragment, projecting the angular vector onto the same observable subspace used by the loss; it had 0.3% more parameters (5,736,898 versus 5,719,970). Both used the covariant orientation map and were trained for 100,000 updates with a global batch of 16 complexes on one GPU, with AdamW, peak learning rate 3×10^{-4} , 2% warmup, cosine decay over the final half and EMA decay 0.999. This budget is much smaller than that of the released model. The analysis plan was fixed before training: the terminal EMA checkpoint of each arm was evaluated without checkpoint selection, the primary endpoint was single-sample internal-validation rollout success (RMSD < 2 Å, 20 late-biased steps, 1 Å translation prior, 8 Å crop, 1,076 complexes), and secondary endpoints were oracle coverage and minimum RMSD in unrefined 100-pose, ten-step banks on Astex and PoseBusters v2. Confidence ranking was not used because the released confidence network consumes features of the released docking network. Rotation equivariance of both heads was verified numerically on validation batches (relative error below 2×10^{-6}). Paired differences use complex-level bootstrap intervals, which describe evaluation rather than training-seed uncertainty.

3.14 Endpoints and statistical analysis

The primary pose endpoint is confidence-selected, symmetry-aware heavy-atom RMSD below 2 Å after complete atom mapping. Secondary endpoints include top- q coverage, the presence of an RMSD-successful candidate among the q highest-ranked candidates; oracle coverage, its presence anywhere in the bank or in the first q generated candidates; and complete-case RMSD summaries. Physical validity uses PoseBusters 0.6.5 in `redock` mode, with a reference-molecule representation adapter for the PhiBench-derived, FoldBench and OpenBind cohorts, as detailed in the [Supplementary Information, Sec. S7](#); pass-all is the conjunction of all 27 non-RMSD checks on the same selected pose [10]. RMSD-PoseBusters success requires both RMSD < 2 Å and PoseBusters pass-all. For Astex and PoseBusters v2, the same official checks were also applied to every raw and refined candidate in all three repeats (235,800 poses). Recomputed checks for each selected and default-ranked candidate were required to reproduce the saved selected-pose evaluation exactly. These labels define a joint oracle, the presence of any candidate with RMSD < 2 Å that passes all checks, and a joint failure partition over all complexes; they are used only for retrospective analysis and never by the selector.

Binary endpoints use the full cohort denominator; continuous summaries use successfully evaluated poses and report their coverage. Error bars are sample standard deviations across the three repeats unless stated otherwise. Supplementary Figs. S9 and S14 use paired 95% bootstrap intervals. Further evaluation details are given in the [Supplementary Information, Sec. S7](#).

3.15 Use of automated tools

OpenAI Codex and Anthropic Claude were used to assist with code review, analysis-script development, manuscript drafting, consistency checks and language editing. The authors reviewed the generated text against the source code and primary literature and retain full responsibility for the manuscript. These tools are not authors.

Data availability

Training data were derived from PLINDER 2024-06/v2 [34]; evaluation structures are available from the Astex Diverse Set, PoseBusters, PhysDock, FoldBench and OpenBind releases cited in the text and remain subject to their original distribution terms. Training-membership lists, per-complex outcomes and the numerical source data for every figure are provided with the repository below.

Code availability

Source code, configuration files, released docking and confidence weights and analysis scripts are available at <https://github.com/eightmm/EFF-Dock>. The version used in this study will be archived with a persistent identifier upon publication.

Author Contributions

J.S.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and writing—original draft. J.L.: conceptualization, methodology, resources, supervision, project administration, and writing—review and editing. Both authors are responsible for the final scientific content.

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Competing Interests

The authors declare no competing financial interest. J.L.’s affiliation with Arontier Co. is disclosed in the author information.

Supplementary Information

Supplementary methods and analyses are provided in the accompanying Supplementary Information.

References

- [1] Oleg Trott and Arthur J. Olson. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, 2010.
- [2] Jerome Eberhardt, Diogo Santos-Martins, Andreas F. Tillack, and Stefano Forli. AutoDock Vina 1.2.0: New docking methods, expanded force field, and python bindings. *Journal of Chemical Information and Modeling*, 61(8):3891–3898, 2021.
- [3] Andrew T. McNutt, Yanjing Li, Rocco Meli, Rishal Aggarwal, and David Ryan Koes. GNINA 1.3: The next increment in molecular docking with deep learning. *Journal of Cheminformatics*, 17:28, 2025.
- [4] Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. DiffDock: Diffusion steps, twists, and turns for molecular docking. In *International Conference on Learning Representations*, 2023.
- [5] Eric Alcaide, Zhifeng Gao, Guolin Ke, Yaqi Li, Linfeng Zhang, Hang Zheng, and Gengmo Zhou. Uni-Mol Docking V2: Towards realistic and accurate binding pose prediction. *arXiv preprint arXiv:2405.11769*, 2024.
- [6] Jintao Zhu, Zhonghui Gu, Jianfeng Pei, and Luhua Lai. DiffBindFR: An SE(3) equivariant network for flexible protein–ligand docking. *Chemical Science*, 15(21):7926–7942, 2024.
- [7] Wei Lu, Jixian Zhang, Weifeng Huang, Ziqiao Zhang, Xiangyu Jia, Zhenyu Wang, Leilei Shi, Chengtao Li, Peter G. Wolynes, Shuangjia Zheng, et al. DynamicBind: Predicting ligand-specific protein–ligand complex structure with a deep equivariant generative model. *Nature Communications*, 15:1071, 2024.
- [8] Alex Morehead and Jianlin Cheng. FlowDock: Geometric flow matching for generative protein–ligand docking and affinity prediction. *Bioinformatics*, 41(Supplement 1):i198–i206, 2025.
- [9] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, John M. Jumper, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024.
- [10] Martin Buttenschon, Garrett M. Morris, and Charlotte M. Deane. PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15:3130–3139, 2024.
- [11] Matthias Rarey, Bernd Kramer, Thomas Lengauer, and Gerhard Klebe. A fast flexible docking method using an incremental construction algorithm. *Journal of Molecular Biology*, 261(3):470–489, 1996.
- [12] Jörg Degen, Christof Wegscheid-Gerlach, Andrea Zaliani, and Matthias Rarey. On the art of compiling and using ‘Drug-Like’ chemical fragment spaces. *ChemMedChem*, 3(10):1503–1507, 2008.
- [13] Alvaro Prat, Leo Zhang, Charlotte M. Deane, Yee Whye Teh, and Garrett M. Morris. SigmaDock: Untwisting molecular docking with fragment-based SE(3) diffusion. In *The Fourteenth International Conference on Learning Representations*, 2026.

- [14] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- [15] Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *The Twelfth International Conference on Learning Representations*, 2024.
- [16] Daria Frolova, Talgat Daulbaev, Egor Sevriugov, Sergei A. Nikolenko, Dmitry N. Ivankov, Ivan Oseledets, and Marina A. Pak. Matcha: Multi-stage riemannian flow matching for accurate and physically valid molecular docking. *arXiv preprint arXiv:2510.14586*, 2026. Version 3.
- [17] Yi-Lun Liao and Tess Smidt. Equiformer: Equivariant graph attention transformer for 3d atomistic graphs. In *International Conference on Learning Representations*, 2023.
- [18] Yufei Huang, Odin Zhang, Lirong Wu, Cheng Tan, Haitao Lin, Zhangyang Gao, Siyuan Li, and Stan Z. Li. Re-Dock: Towards flexible and realistic molecular docking with diffusion bridge. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20474–20489. PMLR, 2024.
- [19] Jaemin Sim and Juyong Lee. BA-Pred and RMSD-Pred: Integrated graph neural network models for accurate protein–ligand binding affinity and binding pose prediction. *Journal of Chemical Information and Modeling*, 66(7):3480–3495, 2026.
- [20] Michael Plainer, Marcella Toth, Simon Dobers, Hannes Stärk, Gabriele Corso, Céline Marquet, and Regina Barzilay. DiffDock-Pocket: Diffusion for pocket-level docking with sidechain flexibility. In *Machine Learning in Structural Biology Workshop, NeurIPS*, 2023.
- [21] J. Henry Broster, Bojana Popovic, Diana Kondinskaia, Charlotte M. Deane, and Fergus Imrie. Teaching diffusion models physics: Reinforcement learning for physically valid diffusion-based docking. *Chemical Science*, 2026.
- [22] Duanhua Cao, Mingan Chen, Runze Zhang, Zhaokun Wang, Manlin Huang, Jie Yu, Xinyu Jiang, Zhehuan Fan, Wei Zhang, Hao Zhou, et al. SurfDock is a surface-informed diffusion generative model for reliable and accurate protein–ligand complex prediction. *Nature Methods*, 22:310–322, 2025.
- [23] Greg Landrum et al. RDKit: Open-source cheminformatics software. <https://www.rdkit.org>, 2024.
- [24] Shuzhe Wang, Jagna Witek, Gregory A. Landrum, and Sereina Riniker. Improving conformer generation for small rings and macrocycles based on distance geometry and experimental torsional-angle preferences. *Journal of Chemical Information and Modeling*, 60(4):2044–2058, 2020.
- [25] Thomas A. Halgren. Merck molecular force field. i. basis, form, scope, parameterization, and performance of MMFF94. *Journal of Computational Chemistry*, 17(5–6):490–519, 1996.
- [26] Ken Shoemake. Uniform random rotations. In David Kirk, editor, *Graphics Gems III*, pages 124–132. Academic Press, 1992.
- [27] Anthony K. Rappé, Carla J. Casewit, K. S. Colwell, William A. Goddard, III, and W. M. Skiff. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *Journal of the American Chemical Society*, 114(25):10024–10035, 1992.
- [28] Sebastian Salentin, Sven Schreiber, V. Joachim Haupt, Melissa F. Adasme, and Michael Schroeder. PLIP: Fully automated protein–ligand interaction profiler. *Nucleic Acids Research*, 43(W1):W443–W447, 2015.
- [29] Georgia B. McGaughey, Marc Gagné, and Anthony K. Rappé. π -stacking interactions: Alive and well in proteins. *Journal of Biological Chemistry*, 273(25):15458–15463, 1998.
- [30] Justin P. Gallivan and Dennis A. Dougherty. Cation– π interactions in structural biology. *Proceedings of the National Academy of Sciences of the United States of America*, 96(17):9459–9464, 1999.

- [31] Gautam R. Desiraju, P. Shing Ho, Lars Kloo, Anthony C. Legon, Roberto Marquardt, Pierangelo Metrangolo, Peter Politzer, Giuseppe Resnati, and Kari Rissanen. Definition of the halogen bond (IUPAC recommendations 2013). *Pure and Applied Chemistry*, 85(8):1711–1713, 2013.
- [32] Ivan Dokmanić, Mile Šikić, and Sanja Tomić. Metals in proteins: Correlation between the metal-ion type, coordination number and the amino-acid residues involved in the coordination. *Acta Crystallographica Section D: Biological Crystallography*, 64(3):257–263, 2008.
- [33] Ruiqi Feng, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. On the guidance of flow matching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 16993–17029. PMLR, 2025.
- [34] Janani Durairaj, Yusuf Adeshina, Zhonglin Cao, Xuejin Zhang, Vladas Oleinikovas, et al. PLINDER: The protein–ligand interactions dataset and evaluation resource. *bioRxiv*, 2024.
- [35] Gabriele Corso, Arthur Deng, Nicholas Polizzi, Regina Barzilay, and Tommi Jaakkola. Deep confident steps to new pockets: Strategies for docking generalization. In *International Conference on Learning Representations*, 2024.
- [36] Alex Morehead, Nabin Giri, Jian Liu, Pawan Neupane, and Jianlin Cheng. Assessing the potential of deep learning for protein–ligand docking. *Nature Machine Intelligence*, 8:32–41, 2026.
- [37] Peter Škrinjar, Jérôme Eberhardt, Gabriel Studer, Gerardo Tauriello, Torsten Schwede, and Janani Durairaj. Evaluating generalization in protein–ligand cofolding methods. *Nature Structural & Molecular Biology*, 33:782–794, 2026.
- [38] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- [39] Michael J. Hartshorn, Marcel L. Verdonk, Gianni Chessari, Suzanne C. Brewerton, Wijnand T. M. Mooij, Paul N. Mortenson, and Christopher W. Murray. Diverse, high-quality test set for the validation of protein–ligand docking performance. *Journal of Medicinal Chemistry*, 50(4):726–741, 2007.
- [40] Kexin Zhang, Yuanyuan Ma, Jiale Yu, Huiting Luo, Jinyu Lin, Yifan Qin, Xiangcheng Li, Qian Jiang, Fang Bai, Jiayi Dou, et al. PhysDock: A physics-guided all-atom diffusion model for protein–ligand complex prediction. *bioRxiv*, 2025.
- [41] Sheng Xu, Qiantai Feng, Lifeng Qiao, Hao Wu, Tao Shen, Yu Cheng, Shuangjia Zheng, and Siqi Sun. Benchmarking all-atom biomolecular structure prediction with FoldBench. *Nature Communications*, 17:442, 2026.
- [42] OpenBind Consortium. OpenBind structure–affinity data release: Enterovirus A71 (EV-A71) / coxsackievirus A16 (CVA16) 2a protease, 2026.